Action Differences, Fluctuations and Operational Bounds

Control results for a research programme on mechanical gaps

navstokgap research working notes

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Abstract

We distinguish a nonzero action scale from a gap in action values, an uncertainty bound and a spectral gap. A constant-force trajectory gives an exact relation between a geometric error area and a fixed-endpoint action difference. The same model excludes an intrinsic positive action-value gap on the unrestricted path space. We give an elementary extension to nondegenerate variations, distinguish Jacobi fields from arbitrary path fluctuations, and test a proposed delta-derivative interpretation using the normalized free quantum kernel. Under explicitly quantum measurement assumptions we derive a finite-resource distinguishability threshold, which is not a universal gap. These are control calculations, not a derivation of quantum mechanics or a claim of novelty. They specify what an extended model would need to change and what remains to be established.

1 Question, notation and evidence status

The motivating question is whether limitations on Newtonian trajectories can explain a nonzero action scale, and whether a useful toy model can isolate the formation of a gap. We write $h = 2\pi\hbar$ for Planck's constants, η for a path variation, and J for a second-variation operator. These are distinct objects. An additional physical field would require its own definition, degrees of freedom and dynamics; it is not supplied by naming η an "h field."

For a path class $\mathcal{P}$ and reference q_*, one possible action-gap question is

$$g_S = \inf\{|S[q] - S[q_*]| : q \in \mathcal{P},\ S[q] \neq S[q_*]\} > 0.$$

The set must be nonempty and the path class, topology, endpoints and duration specified. This is different from the Hamiltonian question $\mathrm{spec}(H) \cap (E_0, E_0 + \Delta) = \varnothing$ for some $\Delta > 0$. For the latter, existence of a ground sector, the operator domain and the meaning of E_0 must also be stated. Neither follows merely from an action parameter's units.

Newton's opening geometric constructions motivate our area comparison [6, 7]. Our use of Hamilton's action and quantum states is modern. The repository records reading coverage by edition and passage. It has not completed a reading of all the Classical Scholia or established that doubts about vanishing areas delayed the 1687 edition. The newly added miscellaneous fragment collection NATP00385 [5] is not itself a complete edition of those scholia. Historical motivation is independent of the proofs below.

2 Constant force and the action-value obstruction

Fix $m, F, T, v_0 > 0$, set $V(y) = -Fy$, and impose $x(0) = y(0) = \dot{y}(0) = 0$, $\dot{x}(0) = v_0$. Newton's equations give

$$x_*(t) = v_0 t, \qquad y_*(t) = \frac{Ft^2}{2m}, \qquad 0 \le t \le T.$$

The kinetic-energy gain, not total-energy change, is $\delta E = F^2T^2/(2m)$. The triangle between initial tangent, endpoint vertical and chord has area $v_0FT^3/(4m)$. The chord–curve lens has one third of this:

$$\mathcal{A}_{\text{lens}} = \frac{v_0FT^3}{12m}.$$

These configuration-space areas are frame dependent; they are not canonical phase-space areas or variances of conjugate observables.

Hold $x = x_*$ fixed and consider the same-endpoint paths $y = y_* + \eta$ with $\eta \in H^1_0(0,T)$, the Sobolev space with square-integrable weak derivative and zero endpoint trace. Define

$$S[x,y] = \int_0^T \left[\frac{m}{2}(\dot{x}^2 + \dot{y}^2) + Fy\right] \mathrm{d}t.$$

The linear variation vanishes by integration by parts and $m\ddot{y}_* = F$:

$$S[x_*, y_* + \eta] - S[x_*, y_*] = \frac{m}{2}\int_0^T \dot{\eta}^2\, \mathrm{d}t. \tag{1}$$

The straight chord $y_{\text{ch}} = FTt/(2m)$ is admissible but is not another solution under the same constant force. Substitution into (1) gives

$$\Delta S_{\text{ch},*} = \frac{F^2T^3}{24m} = \frac{F}{2v_0}\mathcal{A}_{\text{lens}} = \frac{T\delta E}{12}. \tag{2}$$

Adding the same total time derivative $\mathrm{d}G(q,t)/\mathrm{d}t$ to both Lagrangians leaves this difference unchanged because configuration endpoints and duration coincide. A velocity-dependent endpoint function requires additional endpoint data.

Proposition 1 (No unrestricted classical action gap)**.** *For this fixed-endpoint path class the infimum of positive action differences from the classical path is zero, although the classical path is the unique minimizer within the class with fixed horizontal motion.*

Proof. If $\eta \neq 0$ in H^1_0, its derivative cannot vanish almost everywhere, so (1) is positive. For $\eta_a(t) = at(T-t)$ it equals

$$\Delta S(a) = \frac{ma^2T^3}{6}.$$

Any prescribed positive bound is violated by taking a sufficiently small nonzero a. The zero variation alone gives equality in (1). □

Proposition 2 (Local obstruction along a variation)**.** *Let q_a be admissible for all a in an open interval about zero, and suppose $f(a) = S[q_a]$ is twice continuously differentiable there, with $f'(0) = 0$ and $f''(0) = b \neq 0$. Then arbitrarily small nonzero absolute action differences $|S[q_a] - S[q_0]|$ occur. If $b > 0$, these differences can be taken positive without absolute values.*

Proof. Taylor's formula gives $f(a) - f(0) = ba^2/2 + o(a^2)$. For all sufficiently small nonzero a, its magnitude lies between $|b|a^2/4$ and $3|b|a^2/4$ and its sign is the sign of b. It is nonzero and tends to zero. □

Locality here is initially in the parameter a. To conclude that these actions occur in every path-space neighbourhood of q_0, additionally require $q_a \to q_0$ in the specified path topology. The polynomial family above converges in H^1 (and in C^1).

These propositions do not cover discrete path classes, non-smooth restricted actions, different topological sectors or operational equivalence relations. Such changes may be interesting, but must be separately motivated. They also do not resolve existence through singular Kepler collisions: admissible off-shell paths and solutions of an initial-value problem are different sets.

3 Jacobi fields and normalized fluctuation gaps

For $S[q] = \int_0^T (m\dot{q}^2/2 - V(q))\mathrm{d}t$ about a smooth classical solution, the second variation is

$$Q[\eta] = \int_0^T \left[m\dot{\eta}^2 - V''(q_*)\eta^2\right] \mathrm{d}t, \qquad J = -m\frac{\mathrm{d}^2}{\mathrm{d}t^2} - V''(q_*).$$

For smooth bounded $V''(q_*(t))$ take the Dirichlet operator domain $H^2(0,T) \cap H_0^1(0,T)$ in $L^2(0,T)$. A variation through classical solutions satisfies the Jacobi equation $J\eta = 0$; arbitrary off-shell variations do not. For constant force a Jacobi field is $A + Bt$, so fixed zero endpoints force it to vanish. Equation (1) nevertheless admits nonzero fluctuations.

For constant force the Dirichlet eigenfunctions are $\sin(n\pi t/T)$ with eigenvalues $\lambda_n = m(n\pi/T)^2$, $n \geq 1$. Thus

$$Q[\eta] \geq \frac{m\pi^2}{T^2}\|\eta\|_{L^2}^2.$$

This is coercivity relative to a norm, not a smallest nonzero value of Q. Rescaling η rescales Q quadratically. Moreover λ_n has units mass/time^2, while $\|\eta\|_{L^2}^2$ has units length^2 time, producing action. Calling λ_1 a Planck constant would be dimensionally incorrect.

For the oscillator $V = m\omega^2q^2/2$, the corresponding eigenvalues become $m[(n\pi/T)^2 - \omega^2]$. Zero modes occur at $T = n\pi/\omega$; positivity of all modes fails after the first such time. This explicit calculation is a useful next comparison with the quantum oscillator Hamiltonian, whose energy spectrum requires a separate operator and quantization assumptions. Stationary action need not mean a global minimum for arbitrary durations and potentials.

4 What distribution is the oscillatory kernel?

The quantum path-phase input is $\exp(iS/\hbar)$, with time-integrated action, as in the Feynman formulation [2, 3]. A bare exponential is not by itself a normalized distributional approximation to a delta derivative. The free one-dimensional Schrödinger kernel provides a precise control case. For $\hbar, m, \tau > 0$ let

$$K_\tau(z) = \left(\frac{m}{2\pi i\hbar\tau}\right)^{1/2} \exp\left(\frac{imz^2}{2\hbar\tau}\right),$$

with the usual branch $1/\sqrt{i} = e^{-i\pi/4}$ and oscillatory/tempered-distribution interpretation. Use Fourier transform $\widehat{f}(k) = \int e^{-ikz}f(z)\mathrm{d}z$. Equivalently this kernel is defined by

$$\widehat{K_\tau}(k) = \exp\left(-\frac{i\hbar\tau k^2}{2m}\right).$$

For each Schwartz test function, dominated convergence in Fourier space implies

$$K_\tau \longrightarrow \delta \quad (\tau \downarrow 0,\ \hbar > 0 \text{ fixed}). \tag{3}$$

Taylor expansion of the multiplier, using $|e^{-iu} - 1 + iu| \leq u^2/2$ for real u, further gives

$$K_\tau = \delta + \frac{i\hbar\tau}{2m}\delta'' + O(\tau^2) \quad \text{in the sense of pairing with each fixed Schwartz test function.}$$

The remainder estimate follows by integrating k^4 times the absolute value of the transformed test function. The limit is a delta, and the first correction is its second derivative, not its first derivative. The free kernel is even in z. One can obtain δ' by differentiating K_τ with respect to z, but that is a different distribution. A Fourier constraint integral, for example

$\int e^{i\lambda z/\hbar}\mathrm{d}\lambda/(2\pi\hbar) = \delta(z)$, also needs an integration variable and normalization. Substitution of a nonlinear constraint requires further regularity and Jacobian conditions.

This calculation neither constructs a general real-time path-integral measure nor proves a stationary-phase theorem on infinite-dimensional path space. It tests one exact model and keeps the short-time limit separate from semiclassical reasoning and from any proposed physical cutoff.

5 A conditional operational action threshold

Assume the quantum state/measurement framework, a specified positive $\hbar$, and a coherently controlled two-arm system realizing a relative phase $\phi = \Delta S/\hbar$. A global phase on a single state is unobservable. Instead compare two known hypotheses with equal priors:

$$|\psi_0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \qquad |\psi_\phi\rangle = \frac{|0\rangle + e^{i\phi}|1\rangle}{\sqrt{2}}.$$

Allow $N \geq 1$ independent identical copies and any joint binary quantum measurement. This is an idealized hypothesis test, not parameter estimation with an unknown phase and not a claim about all possible resource protocols.

Proposition 3 (Finite-copy bound)**.** *The optimal equal-prior success probability is*

$$P_N(\phi) = \frac{1}{2}\left(1 + \sqrt{1 - \cos^{2N}(\phi/2)}\right). \tag{4}$$

For $1/2 < p \leq 1$, in the first phase lobe $|\phi| \leq \pi$, success at least p is possible if and only if

$$|\Delta S| \geq d_{N,p} := 2\hbar \arccos\left([4p(1-p)]^{1/(2N)}\right). \tag{5}$$

For each fixed $1/2 < p < 1$, $d_{N,p} \to 0$ as $N \to \infty$. For $p = 1$, the finite-copy threshold instead remains $\pi\hbar$.

Proof. Two normalized pure states with overlap magnitude r have density-matrix difference D with eigenvalues $\pm\sqrt{1-r^2}$ on their span (and zero outside). Indeed $\operatorname{tr} D = 0$ and $\operatorname{tr} D^2 = 2(1-r^2)$. A binary measurement effect $0 \leq E \leq I$ gives success $1/2+\operatorname{tr}(ED)/2$, maximized by projecting onto the positive eigenspace. The maximum is $(1+\sqrt{1-r^2})/2$. Here a single-copy overlap has magnitude $|\cos(\phi/2)|$, and tensor products raise it to the Nth power, proving (4). Within the first lobe cosine is nonnegative and monotone in $|\phi|$, so solving $P_N \geq p$ gives (5). For $0 < 4p(1-p) < 1$ its $1/(2N)$ power tends to one. At $p = 1$ it is zero for every finite N, giving the stated exception. $\square$

For example $N = 1$, $p = 3/4$ gives $d_{1,3/4} = \pi\hbar/3$. For fixed $1/2 < p < 1$ the asymptotic threshold is

$$d_{N,p} \sim 2\hbar\sqrt{\frac{-\log(4p(1-p))}{N}}.$$

There is no monotone lower-bound interpretation over all phases: at $\phi = 2\pi n$ the hypotheses coincide again. This model proves a positive operational threshold at fixed resources and accuracy, not a smallest nonzero action in nature. It also assumes, rather than derives, the positive $\hbar$ in the phase rule.

If an apparatus realizes the particular relative action (2), with all control and recombination phases included, its first-lobe threshold would be

$$T \geq \left(\frac{24md_{N,p}}{F^2}\right)^{1/3}.$$

The upper first-lobe condition $F^2T^3/(24m\hbar) \le \pi$ must be retained. The chord is not an alternative trajectory under the original force alone, so this substitution is conditional on a physical implementation, not an experimental prediction for the unmodified projectile problem.

6 Finite propagation and the field-theory companion

Finite c alone does not forbid small variations. A concrete control uses the free relativistic particle, $L = -mc^2\sqrt{1-\dot{q}^2/c^2}$, fixed endpoints zero, and $q_a = at(T-t)$. If $|a|T < c$, the path has a strict speed margin. Relative to the rest path its action excess is positive for $a \neq 0$ and

$$\Delta S(a) = mc^2 \int_0^T \left(1 - \sqrt{1 - a^2(T-2t)^2/c^2}\right) \mathrm{d}t = \frac{ma^2T^3}{6} + O(a^4) \longrightarrow 0.$$

Thus the speed restriction itself supplies no positive action-value gap in this class. This example concerns relativistic particle kinematics, not a complete causal theory of interactions.

Navier–Stokes and Yang–Mills remain complementary comparisons, with their official targets kept intact [1, 4]. A diffusion or Dirichlet coercivity bound can depend on time, domain and normalization; a physical quantum mass gap concerns a different operator and limiting problem. Any future bridge must identify the spaces, physical versus auxiliary time, estimates and limits it preserves. Neither the preceding obstruction nor the operational threshold proves fluid regularity or a four-dimensional quantum-field mass gap.

7 Reproduction, limitations and next experiments

The repository's `make check` verifies the exact polynomial action identities, the oscillator-mode residual, the Fourier expansion coefficient and elementary discrimination cross-checks. These tests supplement the displayed proofs; they are not formal proof certificates. `make papers` rebuilds this note and the companion programme PDF. No Lean proof is supplied in this draft.

The next bounded work is to compare oscillator Hessian and Hamiltonian gaps, free motion on a line and circle, and central-potential collision assumptions. In parallel, primary-source bibliography will compare quantum reconstruction axioms and complete the historical corpus inventory. A successful extension must identify which premise excludes the classical small-amplitude families, or else choose an operational or spectral gap instead and state the change of target. Deriving a theory's form, forcing a nonzero universal scale and determining its measured numerical value are separate tasks.

References

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[2] Richard P. Feynman. Space-time approach to non-relativistic quantum mechanics. *Reviews of Modern Physics*, 20:367–387, 1948. Publisher metadata and abstract checked; no full-paper proof audit.

[3] Richard P. Feynman, Robert B. Leighton, and Matthew Sands. The Feynman lectures on physics, volume ii, chapter 19: The principle of least action. California Institute of Technology, online edition. Section 19–1, quantum discussion following Figure 19–11; consulted 2026-09-05.

[4] Arthur Jaffe and Edward Witten. Quantum Yang–Mills theory. Clay Mathematics Institute, official Millennium problem description. Local source companion records existence and mass-gap requirements.

[5] Isaac Newton. Unarranged fragments, mostly relating to the dispute with Leibniz. The Newton Project, text NATP00385. MS Add. 3968, ff. 594r–619v, Cambridge University Library; electronic transcription completed by Robert Ralley in 2020; consulted 2026-09-05. Manuscript collection has no single composition date assigned here.

[6] Isaac Newton. *Newton's Principia: The Mathematical Principles of Natural Philosophy*. Daniel Adee, New York, 1846. Andrew Motte translation; N. W. Chittenden edition. Definitions, Laws, Book I Sections I–II; local source companion records reading coverage.

[7] Isaac Newton. The mathematical principles of natural philosophy, book i, section i, 2002. Andrew Motte's 1729 translation, edited by David R. Wilkins; local 11-page excerpt.